

Chaitanya Amballa

last updated: June 2026

PhD Student, ECE · University of Illinois Urbana–Champaign (UIUC)

Email: amballa2@illinois.edu

Scholar: [Google Scholar](#)

GitHub: [/achaitu](#)

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Website: achaitanya.web.illinois.edu

Education

University of Illinois Urbana–Champaign

Urbana, IL

Ph.D. in Electrical and Computer Engineering

2022 – present

Advisor: Prof. Romit Roy Choudhury ([SiNRG @ Illinois](#))

Research: generative modeling, spatial inference, diffusion models for inverse problems

University of Illinois Urbana–Champaign

Urbana, IL

M.S. in Electrical and Computer Engineering

2022 – 2025

Advisor: Prof. Romit Roy Choudhury

Indian Institute of Technology Tirupati

Tirupati, India

B.Tech. in Electrical Engineering

2015 – 2019

Advisor: Prof. K. P. Naveen

Honors & Awards

Shun Lien Chuang Memorial Award for Excellence in Graduate Education

2026

Best Poster Award, HotMobile 2026

2026

Qualcomm Graduate Award

2025

Best Bowler Award, Illinois Premier League

2022

Silver Medals, Intra IIT Table Tennis and Cricket

2018

INSPIRE Scholarship

2015

IMMO Silver Medal

2013

Sai Pratibha Puraskar

2013

NSO & NCO Gold Medals

2007 & 2008

Preprints

[1] Discrete Langevin-Inspired Posterior Sampling



[Chaitanya Amballa](#)^{*}, Sattwik Basu^{*}, Jorge Vančo Sampedro, Romit Roy Choudhury

(* = equal contribution)

[2] Can Sparse Walking Patterns Disclose Spatial Maps?



[Chaitanya Amballa](#)^{*}, Sattwik Basu^{*}, Mrudula Athi, Mohamed Mansour, Mehmet Ergezer, Wontak Kim, Romit Roy Choudhury

(* = equal contribution)

[3] Learning Energy-based Variational Latent Prior for VAEs



Debottam Dutta, [Chaitanya Amballa](#), Zhongweiyang Xu, Yu-Lin Wei, Romit Roy Choudhury

Publications

[1] Explicit Context-Driven Neural Acoustic Modeling for High-Fidelity RIR Generation



Chen Si, Qianyi Wu, [Chaitanya Amballa](#), Romit Roy Choudhury

Interspeech, 2026

- [2] **Blind Audio Restoration using Contrastive Diffusion Guidance** [\[project page\]](#) 
 Sattwik Basu*, [Chaitanya Amballa](#)*, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi,
 Romit Roy Choudhury
 (* = equal contribution)
ICML Workshop on ML for Audio, 2026
- [3] **Contrastive Diffusion Guidance for Spatial Inverse Problems** [\[project page\]](#) 
[Chaitanya Amballa](#)*, Sattwik Basu*, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi,
 Romit Roy Choudhury
 (* = equal contribution)
ICLR, 2026
- [4] **Inferring Indoor Layouts using Audio** 
 Zhijian Yang, Sattwik Basu*, [Chaitanya Amballa](#)*, Debottam Dutta, Srihari Nelakuditi, Romit Roy
 Choudhury
HotMobile, 2026
- [5] **Can NeRFs See without Cameras?** [\[project page\]](#) 
[Chaitanya Amballa](#), Yu-Lin Wei, Sattwik Basu, Zhijian Yang, Mehmet Ergezer, Romit Roy Choudhury
NeurIPS, 2025
- [6] **Identifying Near-Optimal Decisions in Linear-in-Parameter Bandit Models with Continuous Decision Sets** 
 Sanjay P. Bhat, [Chaitanya Amballa](#)
UAI, 2022
- [7] **Computing an Efficient Exploration Basis for Learning with Univariate Polynomial Features** 
[Chaitanya Amballa](#), Manu K. Gupta, Sanjay P. Bhat
AAAI, 2021
- [8] **Coexistence of LTE-Unlicensed and WiFi: A Reinforcement Learning Framework** 
 K. P. Naveen, [Chaitanya Amballa](#)
COMSNETS, 2021
- [9] **Learning Algorithms for Dynamic Pricing: A Comparative Study** 
[Chaitanya Amballa](#), Narendhar Gugulothu, Manu K. Gupta, Sanjay P. Bhat
ICML Workshop on Real World Experiment Design and Active Learning, 2020
- [10] **Coexistence of LTE-Unlicensed and WiFi: Optimization and Game-Theoretic Frameworks** 
[Chaitanya Amballa](#), K. P. Naveen
SPCOM, 2020

Research
Projects

Human Body Pose Estimation with No Fine-Tuning [\[project page\]](#)

Oct 2025 – Dec 2025

Advisor: Dr. Zhizhen Zhao

Diffusion and flow matching for HMD-based pose estimation; posterior sampling for generalization across body shapes

	RIR Generation and Geometry Estimation Advisor: Dr. Minje Kim <i>RIR generation using diffusion models; optimizing for unknown source position using Alexa measurements</i>	Feb 2025 – May 2025
	Click2Hear: Spatial Audio Separation using Visual Cues [project page] Advisor: Dr. Saurabh Gupta <i>Unsupervised source separation; binaural audio from video with mono audio</i>	Aug 2022 – Dec 2022
	Co-existence of LTE-Unlicensed and WiFi Advisor: Dr. K. P. Naveen <i>Optimization, game-theoretic, and Q-learning frameworks for spectrum coexistence</i>	Aug 2018 – May 2019
	Speech Enhancement Advisor: Dr. Rama Krishna Sai Gorthi <i>Convolutional autoencoder networks for real-time speech denoising</i>	Aug 2018 – Dec 2018
	Blood Cell Classification and Disease Diagnosis Advisors: Dr. Rama Krishna Sai Gorthi & Dr. Subrahmanyam Gorthi <i>CNN-based classification of blood cells and disease detection from smear images</i>	Sep 2017 – Jun 2018
Industry Experience	Applied Scientist Intern Amazon	May 2026 – Aug 2026 Cambridge, MA
	Applied Scientist Intern Amazon	May 2025 – Aug 2025 Cambridge, MA
	Researcher TCS Research Sequential decision making, stochastic linear bandits, Thompson sampling, regret analysis under Dr. Sanjay P. Bhat	Jul 2019 – Jul 2022 Hyderabad, India
	Project Associate IIT Tirupati	Jun 2019 – Jul 2019 Tirupati, India
	Research Intern Lekha Wireless Solutions	May 2018 – Jul 2018 Bengaluru, India
Teaching	Teaching Assistant ECE 498/598: Deep Generative Models <i>Office hours, grading, and student support</i>	University of Illinois Urbana–Champaign Spring 2026
	Teaching Assistant ECE 434: Real-World Algorithms for IoT and Data Science <i>Office hours, grading, and student support</i>	University of Illinois Urbana–Champaign Spring 2024, Spring 2025
Skills	Generative Modeling: Diffusion Models, NeRFs, Flow Matching, VAEs Frameworks: PyTorch, TensorFlow, HuggingFace Diffusers/Transformers Tooling: Slurm, Weights & Biases, Git, Streamlit, $\LaTeX$ Programming: Python, C/C++, MATLAB, SQL	

Selected Coursework	ECE 598: Deep Generative Models	
	ECE 598: Generative Modeling with Diffusion and Flow Matching Models	
	CS 598: Generative Models for Audio	
	ECE 549: Computer Vision	
	ECE 543: Statistical Learning Theory	
	ECE 586: MDPs and Reinforcement Learning	
	ECE 534: Random Processes	
	ECE 490: Introduction to Optimization	
Thesis	[1] M.S. Thesis	
	Chaitanya Amballa	
	<i>M.S. in Electrical and Computer Engineering, UIUC, 2025</i>	
	[2] B.Tech. Thesis	
	Chaitanya Amballa	
	<i>B.Tech. in Electrical Engineering, IIT Tirupati, 2019</i>	
Leadership	House Captain, Team “Sivaliks”	IIT Tirupati, 2016–2017
	Member, Technical Committee	IIT Tirupati, 2017–2019
	Electrical Engineering Department Sports Representative	IIT Tirupati, 2017–2018